LANGUAGE LEARNING WEBAPP WITH QUIZZES

AI23431 - WEB TECHNOLOGY AND MOBILE APPLICATION

MINI PROJECT REPORT

Submitted by

KARTHIK SAH E (241501080)
KING PAVIYON MANOVA J (241501086)

in partial fulfillment of the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE MACHINE LEARNING

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI
An Autonomous Institute

CHENNAI - 602 105

APRIL 2026

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this Mini Project Report titled "LANGUAGE LEARNING APP WITH QUIZZES" is the Bonafide work of KARTHIK SAH E (241501080) and KING PAVIYON MANOVA J (241501086) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

Dr. A. RAJASEKARAN
Professor,
Department of Artificial
Intelligence and
Machine Learning,
Rajalakshmi Engineering
College,
Chennai - 602105.

Submitted to Project Viva-Voce Examination held on ....

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavour to put forth this report. Our sincere thanks to our Chairman **Mr. S. MEGANATHAN, B.E, F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.,** and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.,** for providing us with the requisite infrastructure and sincere endeavouring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.,** our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. K. SEKAR, Ph.D.,** Professor and Head of the Department of Artificial Intelligence and Machine Learning for his guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide,

Dr. A. RAJASEKARAN, Department of Artificial Intelligence and Machine Learning for his useful tips during our review to build our project.

KARTHIK SAH E (241501080)
KING PAVIYON MANOVA J (241501086)

DEPARTMENT VISION

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION

- To produce globally competent professionals, motivated to learn the emerging technologies and to be innovative in solving real world problems.
- To promote research activities amongst the students and the members of faculty that could benefit the society.
- To impart moral and ethical values in their profession.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOS)

PEO 1: To equip students with essential background in computer science with emphasis on Artificial Intelligence, Machine Learning, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications using emerging technologies.

PEO 3: To encourage research and innovative project development in the field of Artificial Intelligence, Machine Learning, Deep Learning, Networking, Security, Web development, Data Science and also emerging technologies for social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry requirements.

PROGRAMME OUTCOMES (POS)

PO 1: Engineering Knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem Analysis: Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

PO 5: Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and Sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOS)

A graduate of the Artificial Intelligence and Machine Learning Program will demonstrate:

PSO 1: Foundation Skills: Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, AI, machine learning, deep learning, data science, and networking.

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational tasks, model real world problems using appropriate AI and ML algorithms. To understand the standard practices and strategies in project development, using open-ended programming environments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible AI and ML professional.

COURSE OBJECTIVES

To provide foundational knowledge and practical skills in creating and structuring web pages using HTML, enabling students to build accessible and well-organized websites.

- To understand and practice Embedded Dynamic Scripting on Client-side Internet Programming.
- To implement Server-Side Scripting.
- To facilitate students to understand Android Application Design.
- To help students to gain a basic understanding of Android application development.

COURSE OUTCOMES

CO 1: Upon completing the course, students will be able to design and develop basic web pages with proper structure and semantic elements, ensuring accessibility and functionality.

CO 2: Design and implement dynamic web page with validation using JavaScript objects and by applying different event handling mechanism.

CO 3: Design and implement simple webpage to learn JSP and Servlet.

CO 4: Design and implement simple Application Design.

CO 5: Identify various concepts of mobile programming that make it unique from programming for other platforms.

CO - PO — PSO MATRICES OF COURSE

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
AI23431.01	3	3	3	3	3	3	2	2	3	2	1	3	3	3	2
AI23431.02	3	3	3	3	3	3	2	2	3	2	1	1	3	3	2
AI23431.03	3	3	3	3	3	3	2	2	3	2	2	2	3	3	3
AI23431.04	3	3	3	3	3	3	2	2	3	2	2	3	3	3	3
AI23431.05	3	3	3	3	3	3	2	2	3	2	3	3	3	3	3
Average	**3**	**3**	**3**	**3**	**3**	**3**	**2**	**2**	**3**	**2**	**1.8**	**2.1**	**3**	**3**	**2.4**

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
SDG-4	**Quality Education**	The Language Learning App with Quizzes promotes inclusive and equitable quality education by providing learners with structured, self-paced vocabulary lessons, timed quizzes, and immediate feedback, enabling students to gain practical language skills and become industry-ready professionals.
SDG-8	**Decent Work and Economic Growth**	The application equips students with multilingual communication skills that improve their employability and career prospects in a globally connected job market. By offering language training through an interactive digital platform, it contributes to personal economic development and growth.
SDG-9	**Industry, Innovation and Infrastructure**	The project promotes innovation by applying modern web technologies including PHP, MySQL, JavaScript, HTML5, and CSS3 to build a scalable and accessible digital platform for language education, contributing to digital infrastructure and technological advancement.
SDG-10	**Reduced Inequalities**	By offering a free, browser-based language learning solution with no subscription or installation required, the application bridges the digital and educational divide and ensures equal access to multilingual learning resources for students from all backgrounds.

Note: Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low)   2: Moderate (Medium)   3: Substantial (High)

If there is no correlation, put "-"

ABSTRACT

The Language Learning App with Quizzes (LLAQ) is a web-based interactive platform designed to help users learn multiple foreign languages through structured lessons, vocabulary practice, and timed quiz assessments. The application addresses the growing demand for affordable, self-paced, and engaging language education by providing a gamified learning environment that motivates learners through points, levels, and a competitive leaderboard system.

The system supports eight languages including English, French, German, Japanese, Korean, Mandarin, Hindi, and Tamil. Each language is organized into progressive levels, with each level containing a vocabulary lesson followed by a timed multiple-choice quiz. Users earn points upon successful quiz completion, which contributes to their ranking on a global leaderboard visible to all registered users.

The application implements role-based user authentication, session management for tracking login state, and cookie-based preference storage. The frontend is developed using HTML, CSS, and JavaScript with a modern dark-themed UI, while the backend is built using PHP and MySQL for centralized data storage and management. The system demonstrates practical implementation of key web technology concepts including HTTP methods, session handling, and cookie management.

Overall, the Language Learning App with Quizzes promotes accessible language education, improves vocabulary retention through interactive quizzes, and encourages consistent learning through its gamification features. The system is lightweight, browser-accessible, and designed for deployment in educational institutions and individual use alike.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
LLAQ	Language Learning App with Quizzes
DB	Database
SQL	Structured Query Language
API	Application Programming Interface
UI	User Interface
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
HTTP	HyperText Transfer Protocol
PHP	Hypertext Preprocessor
CRUD	Create, Read, Update, Delete
MCQ	Multiple Choice Question
SMTP	Simple Mail Transfer Protocol
REST	Representational State Transfer

CHAPTER 1

INTRODUCTION

1.1 General

Language learning is a critical skill in today's interconnected world, enabling communication across cultures, improving career prospects, and enhancing cognitive abilities. However, traditional language learning methods such as classroom instruction and printed textbooks are often expensive, inflexible, and lack the interactive elements that modern learners expect. Digital language learning platforms have emerged as an effective alternative, offering self-paced learning, immediate feedback, and engaging interactive content.

The Language Learning App with Quizzes (LangLearn) is a web-based application that provides users with a structured and gamified approach to learning foreign languages. The system organizes learning content into language-specific levels, each containing a vocabulary lesson and a corresponding timed quiz. Users accumulate points by completing quizzes successfully, and their scores are reflected on a competitive leaderboard that motivates continuous learning.

The application is built using modern web technologies including HTML, CSS, JavaScript, PHP, and MySQL. It features a clean and visually appealing dark-themed interface that makes learning enjoyable and minimizes cognitive fatigue. The system supports eight languages and is designed to be easily extended with additional languages and content.

1.2 Need for Language Learning App with Quizzes

Many language learners struggle to maintain motivation and track their progress over time. Traditional language courses are often costly and require fixed schedules, while purely text-based learning lacks the engagement needed for long-term retention. There is a clear need for an accessible, interactive, and self-paced language learning platform that combines structured lessons with quiz-based assessment and motivational features.

A web-based Language Learning App with Quizzes addresses these needs by providing learners with immediate feedback through quizzes, a sense of achievement through points and level completion, and healthy competition through a leaderboard. The platform makes quality language education accessible to anyone with a browser, without the need for specialized software or costly subscriptions.

1.3 Scope of the System

The Language Learning App with Quizzes is developed to provide a complete digital platform for self-paced language learning. The system supports user registration and login with session-based authentication, allowing each learner to maintain their personal progress and points across sessions.

Users can choose from eight available languages and progress through structured levels, each consisting of a vocabulary lesson followed by a timed multiple-choice quiz. The system enforces sequential level progression, locking higher levels until the previous level's quiz is completed. A global leaderboard displays all registered users ranked by their total points, fostering a competitive and motivating learning environment.

The scope also includes the demonstration of web technology concepts such as HTTP session management for tracking authentication state, cookie-based storage of user preferences, and REST API interactions for data retrieval. The system can be deployed in educational institutions, language training centres, and for individual self-study.

1.4 Organization of the Report

Chapter 1 provides the introduction to the project, including the general overview, need for the system, and scope of the project.

Chapter 2 presents the literature survey discussing existing language learning platforms and related work in this domain.

Chapter 3 explains the problem formulation and objectives of the proposed system.

Chapter 4 describes the system design, including architecture and system requirements.

Chapter 5 discusses the implementation of the system and explains the different modules with screenshots.

Chapter 6 presents the results and performance evaluation of the system.

Chapter 7 concludes the project and discusses possible future enhancements.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Language learning technology has evolved significantly over the past two decades, transitioning from static computer-aided language learning (CALL) tools to fully interactive, gamified web and mobile applications. Research in this area highlights the effectiveness of spaced repetition, immediate feedback, and social motivation in improving language acquisition outcomes. This chapter reviews existing language learning platforms and relevant research works that informed the design of the Language Learning App with Quizzes.

Recent studies emphasize the growing adoption of web-based and mobile language learning platforms, quiz-based assessment systems, and gamification techniques such as points, levels, and leaderboards. These approaches aim to increase learner engagement, improve vocabulary retention, and sustain long-term motivation through structured and interactive learning experiences.

2.2 Existing Language Learning Platforms

Several popular language learning platforms have shaped the digital language education landscape. Duolingo, one of the most widely used language learning applications, employs gamification through experience points, streaks, and leaderboards to motivate daily practice. Its lesson structure alternates between vocabulary introduction and interactive exercises, demonstrating the effectiveness of combining learning content with immediate quiz-based assessment.

Memrise uses spaced repetition and mnemonic techniques to help learners retain vocabulary, while Babbel focuses on conversational lessons with structured grammar explanations. These platforms demonstrate the value of organizing language content into progressive levels and providing real-time feedback on learner performance. However, most commercial platforms require subscriptions and are not easily customizable for institutional use.

2.3 Related Work

Research on gamification in education consistently shows that elements such as points, badges, levels, and leaderboards significantly improve learner engagement and completion rates in online learning environments. Studies on web-based quiz systems demonstrate that timed multiple-choice questions are effective for assessing vocabulary knowledge and reinforcing learning through immediate correct-answer feedback.

Research on session management in web applications highlights the importance of server-side session storage for maintaining user authentication state across multiple page visits. Cookie-based preference management has been studied as a method for persisting user settings such as theme preferences and learning history. Studies on leaderboard systems in educational platforms confirm that competitive ranking motivates learners to increase their practice frequency and strive for higher scores.

2.4 Summary

From the literature survey, it is evident that effective language learning platforms combine structured vocabulary lessons with quiz-based assessment, gamification elements, and social motivation features. Existing commercial platforms demonstrate the value of these approaches but often lack customizability for specific institutional contexts. The proposed Language Learning App with Quizzes addresses these gaps by providing a locally deployable, web-based platform built with open-source technologies, offering structured multilingual lessons, timed quizzes, point-based progression, and a competitive leaderboard system accessible without subscriptions.

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

Language learners, particularly students in educational institutions, often lack access to affordable, structured, and interactive language learning resources. While commercial platforms exist, they typically require paid subscriptions, are not tailored to institutional curricula, and do not support deployment within a controlled educational environment.

Traditional classroom-based language instruction is limited by fixed schedules and a lack of personalized pacing. Students who wish to practice languages outside of class hours have few structured options that provide immediate feedback and track their progress. Additionally, there is no platform that combines vocabulary lessons, quiz assessments, point-based progression, and competitive leaderboards in a single, locally deployable web application.

Therefore, there is a need for a web-based Language Learning App with Quizzes that provides self-paced, structured language lessons, timed quiz assessments with immediate feedback, a gamified progression system with points and levels, and a competitive leaderboard — all accessible through a standard web browser without requiring any installation or subscription.

3.2 Objectives of the System

The main objective of the Language Learning App with Quizzes is to develop an interactive digital platform that makes language learning engaging, structured, and accessible. The specific objectives include:

- To support multiple languages with organized, level-based learning content including vocabulary lessons and quizzes.
- To provide secure user registration and login with session-based authentication.
- To implement a progressive level system that unlocks higher levels upon successful quiz completion.
- To deliver timed multiple-choice quizzes that assess vocabulary knowledge and provide immediate feedback.
- To award points for quiz completion and maintain cumulative user scores in the database.
- To display a global leaderboard ranking all registered users by their total points.

- To implement HTTP session management for tracking user authentication state across pages.
- To use cookies for storing user preferences and session data.
- To provide a visually appealing and responsive user interface suitable for desktop and mobile browsers.

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design defines the overall structure, components, and interactions of the application. A well-planned design ensures efficiency, maintainability, and scalability. The Language Learning App with Quizzes is designed as a web-based application using a three-tier architecture that separates the user interface, application logic, and data storage into distinct, manageable layers.

4.2 System Architecture

The system follows a three-layer architecture:

Presentation Layer:

This layer represents the user interface through which learners interact with the application. It includes the login and registration pages, the user dashboard displaying progress statistics, the language selection page, the level listing page, the vocabulary lesson page, the quiz page, the quiz results page, and the leaderboard page. The frontend is developed using HTML, CSS, and JavaScript with a modern dark-themed design.

Application Layer:

This layer contains the business logic of the system. It is implemented using PHP and handles user authentication and session management, language and level data retrieval, vocabulary lesson delivery, quiz question loading and answer validation, point calculation and score storage, and leaderboard data aggregation.

Database Layer:

This layer stores all system data using MySQL. Key tables include: Users (user credentials, points, and session data), Languages (language names and codes), Levels (level titles, lesson content per language), Questions (quiz questions with answer options and correct answers per level), and Progress (user-level completion records and scores).

4.3 Database Design

The database is structured to efficiently support all system functionalities. The Users table stores user ID, username, email, hashed password, total points, and registration date. The Languages table stores language ID, name, and country code. The Levels table stores level ID, language ID, level number, title, and lesson content. The Questions table stores question ID, level ID, question text, four answer options, and the correct answer index. The Progress table records user ID, level ID, completion status, score, and completion timestamp.

4.4 System Requirements

4.4.1 Hardware Requirements

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Storage: 500 GB Hard Disk or SSD
- Input Devices: Keyboard and Mouse
- Internet Connection

4.4.2 Software Requirements

- Operating System: Windows 10 or later / Ubuntu Linux
- Frontend Technologies: HTML5, CSS3, JavaScript
- Backend Technology: PHP 8.x
- Database: MySQL 8.x
- Local Server: XAMPP / WAMP
- Development Environment: Visual Studio Code
- Web Browser: Google Chrome or any modern browser

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

The Language Learning App with Quizzes is implemented using a modular web-based approach, where each functional component of the system is developed as an independent module. This improves maintainability and allows future enhancements to be added without disrupting existing functionality.

The frontend is developed using HTML5, CSS3, and JavaScript, providing a modern dark-themed responsive interface. The backend is implemented using PHP, handling user authentication, session management, language and level data retrieval, quiz processing, point calculation, and leaderboard management. MySQL stores all user data, language content, quiz questions, and progress records centrally.

The system implements HTTP session management to maintain user authentication state across pages, and browser cookies to store user preferences. The modular architecture ensures clean separation of concerns and allows the system to be easily extended with new languages, levels, and quiz content.

5.2 Module Descriptions

The Language Learning App with Quizzes consists of the following integrated modules:

5.2.1 Login and Authentication Module

The Login and Authentication Module serves as the secure entry point to the LangLearn platform. Users can register a new account by providing their name, email, and password, or log in using existing credentials. The system validates input fields, hashes passwords before storage, and creates a server-side session upon successful login.

The login page displays the LangLearn branding with the tagline "Master languages like never before!" and provides fields for email and password entry. New users are directed to the sign-up page through the registration link. Session cookies are set upon login to maintain authentication state across all pages of the application.

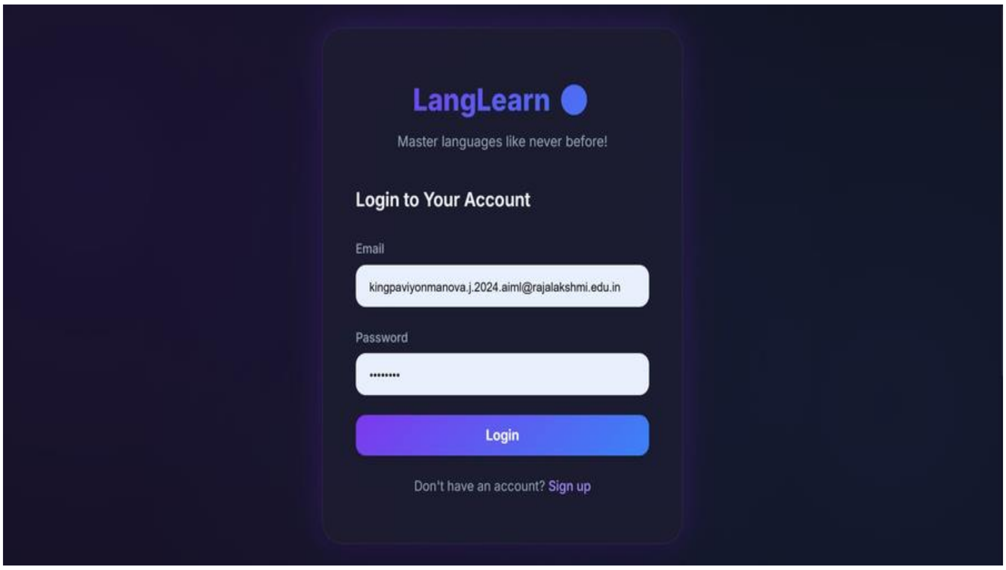

Fig 5.1 - Login and Authentication Page

5.2.2 Dashboard Module

The Dashboard Module is the main home page of the application after login. It provides a personalized welcome message with the user's name and displays key learning statistics in a card-based layout. The statistics shown include Total Points earned, number of Languages Started, number of Levels Completed, and the total number of Languages Available on the platform.

The sidebar navigation includes links to the Dashboard and Leaderboard, along with the user's profile showing their name and current points. A language selection section at the bottom of the dashboard allows users to directly choose a language and begin or continue their learning journey. The dashboard provides a quick overview of the user's overall progress at a glance.

Fig 5.2 - User Dashboard with Learning Statistics

5.2.3 Language Selection Module

The Language Selection Module displays all available languages on the platform in a responsive grid layout. Each language is represented by a card showing the country code abbreviation, the full language name, the number of levels completed out of the total available levels, the completion percentage, and a visual progress bar.

The platform currently supports eight languages: English (GB), French (FR), German (DE), Japanese (JP), Korean (KR), Mandarin (CN), Hindi (IN), and Tamil (IN). Languages in which the user has already started learning display their progress, while untouched languages show 0% completion. Clicking on a language card navigates the user to the levels page for that language.

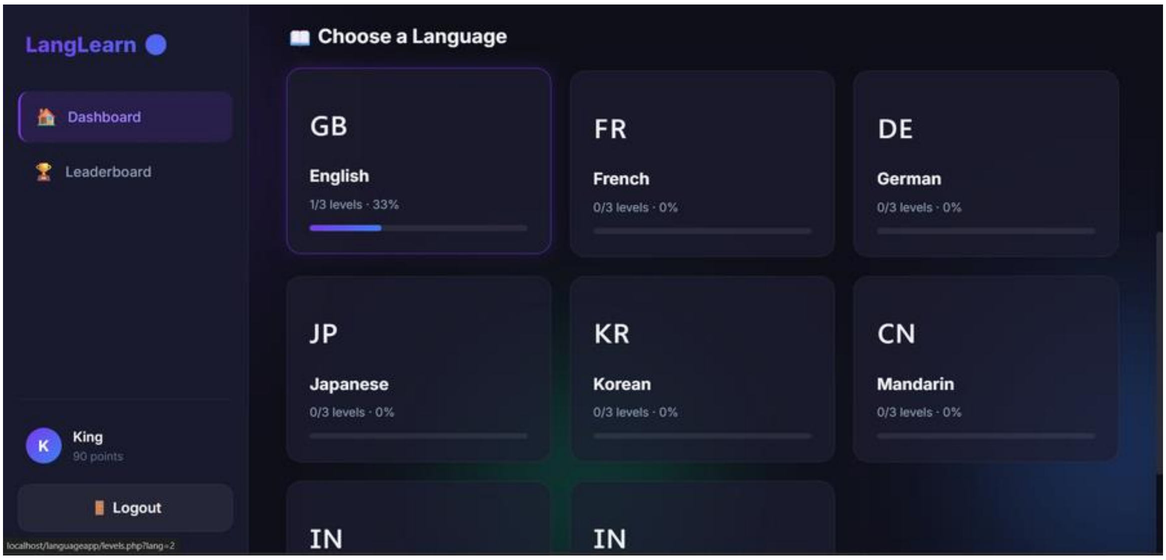

Fig 5.3 - Language Selection Page with Progress Indicators

5.2.4 Level and Lesson Module

The Level and Lesson Module displays all available levels for a selected language in a sequential list. Each level entry shows the level number, lesson topic title, and a status indicator. Completed levels are marked with a green "Completed" badge, the current available level shows a "Start" button, and future locked levels display a lock icon indicating they cannot be accessed until the previous level is completed.

A progress bar at the top of the page shows the overall completion percentage for the selected language. For the English language, the three available levels are: Level 1 - Basic Greetings, Level 2 - Common Phrases, and Level 3 - Travel and Directions. This sequential progression ensures that learners build vocabulary systematically before advancing to more complex content.

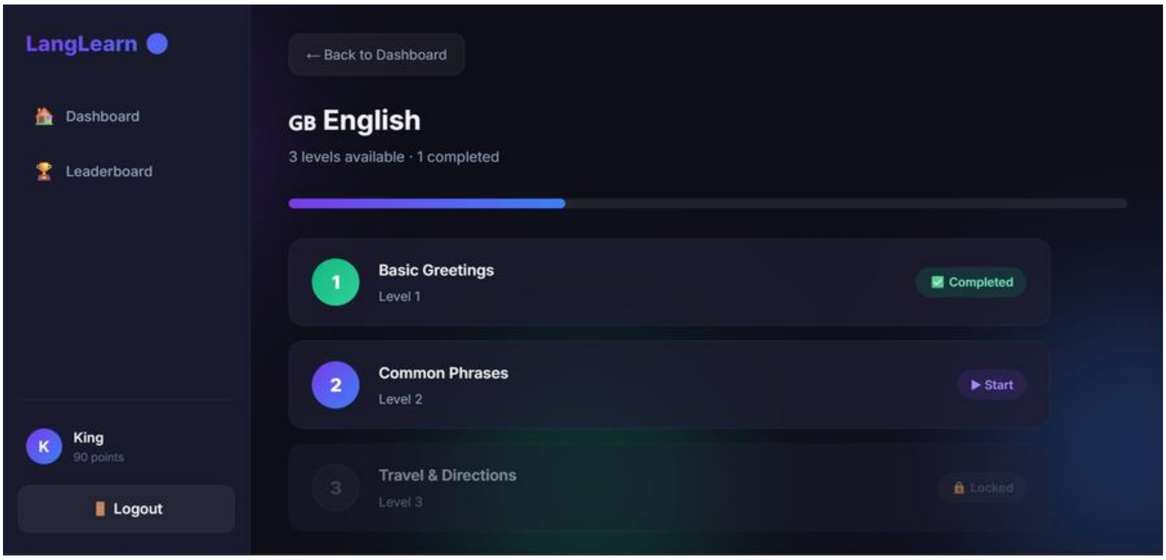

Fig 5.4 - Language Level Listing with Completion Status

5.2.5 Vocabulary Learning Module

The Vocabulary Learning Module presents the lesson content for each level before the user proceeds to the quiz. Each vocabulary card displays the target word or phrase in the selected language, its meaning or translation, a usage context description, and an example sentence shown in a styled quote block. This format ensures that learners understand not just the meaning but also the practical usage of each vocabulary item.

For example, in the English Level 2 - Common Phrases lesson, vocabulary items such as "Excuse me" (Getting attention politely) and "How much?" (Asking about price) are presented with contextual example sentences. Users scroll through all vocabulary items before proceeding to the quiz at the bottom of the lesson page.

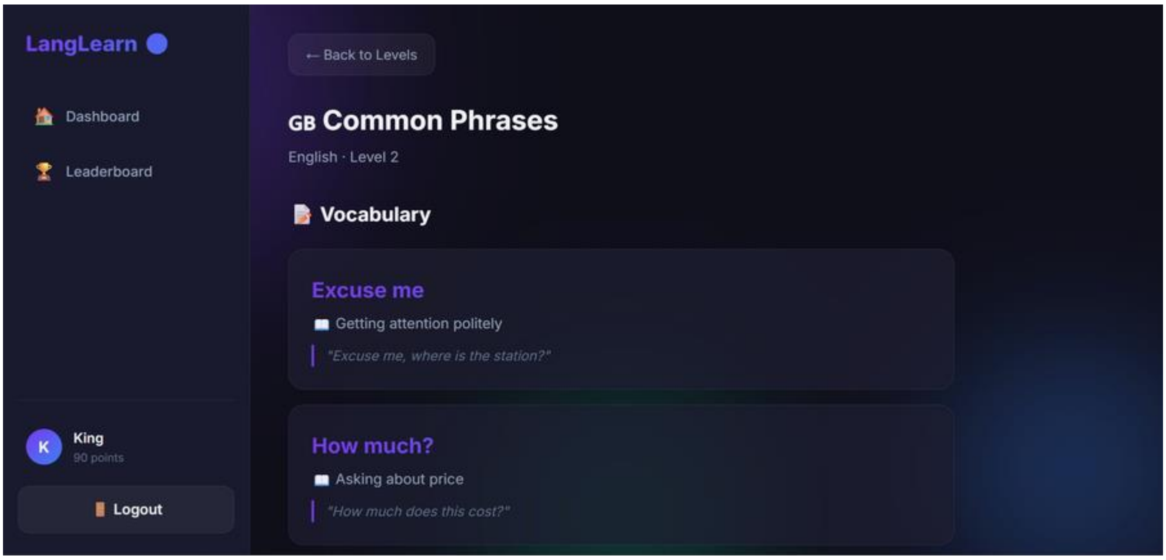

Fig 5.5 - Vocabulary Lesson Page with Example Sentences

5.2.6 Quiz Module

The Quiz Module is the core assessment component of the system. After completing the vocabulary lesson, users take a timed multiple-choice quiz to test their understanding of the lesson content. The quiz page displays the quiz title, language, level, and total number of questions. A countdown timer in the top-right corner adds urgency and encourages focused answering.

Each question is displayed one at a time with four answer options labeled A, B, C, and D. Users select an answer by clicking the corresponding option. For the English Level 2 quiz titled "Common Phrases", sample questions include "'Excuse me' is used to?" with options such as "Get attention politely", "Say goodbye", "Order food", and "Ask directions". The quiz consists of 5 questions per level and awards points based on the number of correct answers.

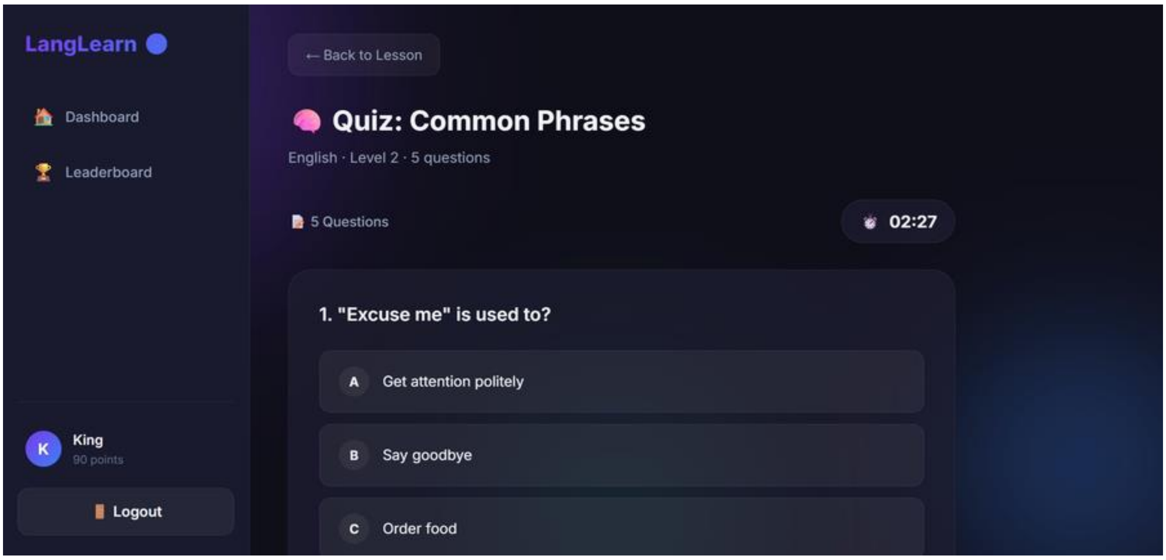

Fig 5.6 - Quiz Page with Timed Multiple Choice Questions

5.2.7 Quiz Result and Points Module

The Quiz Result Module displays the outcome of the completed quiz immediately after the user submits their answers. The result page shows a celebratory message based on performance (such as "Excellent! You nailed it!" for high scores), the number of correct answers out of the total questions (e.g., 4/5), and the total points earned for that attempt displayed in green (e.g., +40 points earned).

The earned points are automatically added to the user's total cumulative score, which is updated in the database and reflected in the sidebar, dashboard statistics, and leaderboard. Three navigation buttons at the bottom allow the user to Retry Quiz, view All Levels, or return to the Dashboard. If the quiz is passed, the next level is automatically unlocked for the user.

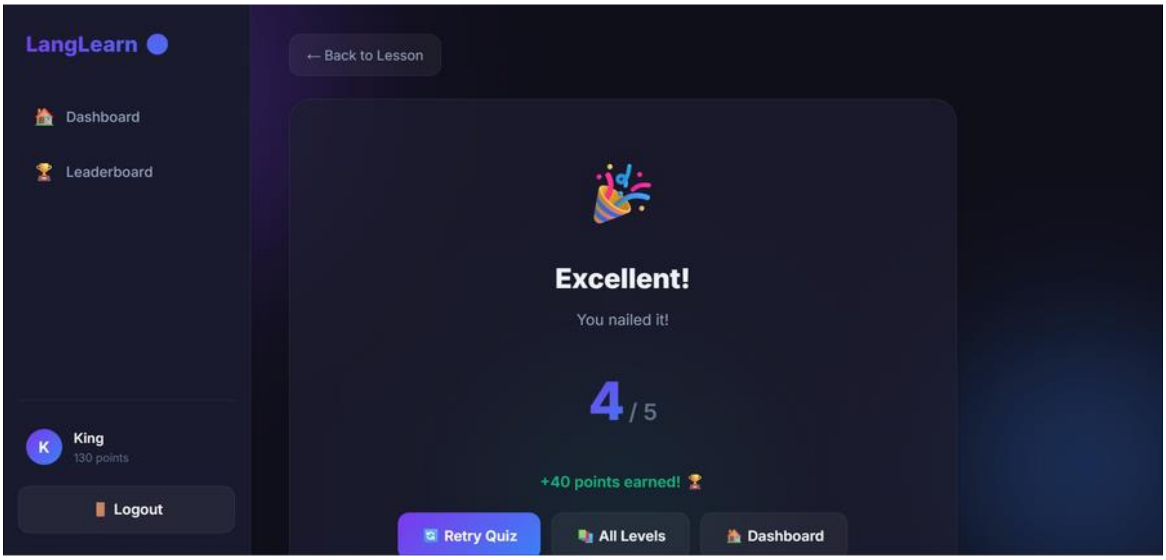

Fig 5.7 - Quiz Result Page with Score and Points Earned

5.2.8 Leaderboard Module

The Leaderboard Module displays a competitive ranking of all registered users sorted by their total accumulated points. At the top of the leaderboard page, the current user's rank, total points, and the number of total players on the platform are shown as summary statistics. The main leaderboard table lists each player with their rank badge, username, and total points.

The current user's entry is highlighted to make it easy to identify their position relative to others. Gold, silver, and bronze medals are displayed for the top three ranked players. This competitive element encourages users to complete more quizzes and improve their scores to climb the rankings. The leaderboard updates in real time as users complete quizzes and earn new points.

Fig 5.8 - Leaderboard with User Rankings and Points

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Results

The Language Learning App with Quizzes was successfully developed and tested as a fully functional web-based platform. The system demonstrated effective performance across all implemented modules including user registration and login, personalized dashboard with statistics, multi-language selection with progress tracking, sequential level progression with lock/unlock logic, vocabulary lesson delivery, timed multiple-choice quiz assessment, points-based scoring, quiz result display, and a competitive leaderboard.

During testing, the system accurately processed user registrations and logins with proper session creation. The dashboard correctly displayed real-time statistics reflecting the user's total points, languages started, and levels completed. The language selection page accurately showed individual language progress with percentage bars. The level listing correctly enforced sequential progression by locking and unlocking levels based on completion status.

Vocabulary lessons were delivered with complete content including word definitions and example sentences. The quiz module accurately displayed timed questions and accepted user responses. Points were correctly calculated and immediately reflected in the user's profile and the leaderboard. All test cases passed, confirming that the system is reliable and ready for deployment.

6.2 Performance Evaluation

The performance of the Language Learning App with Quizzes was evaluated based on usability, accuracy, response time, and reliability. The system provides a modern, intuitive dark-themed interface that is visually appealing and easy to navigate for learners of all technical backgrounds. All major operations including login, language selection, lesson viewing, quiz submission, and leaderboard display were completed within acceptable response times.

The PHP backend efficiently handles user requests, session operations, and database queries. The MySQL database maintains data integrity through relational constraints and proper indexing, allowing fast retrieval of quiz questions, user progress, and leaderboard data. The modular design ensures that individual components can be updated independently without affecting the rest of the system.

6.3 System Testing

The system was tested through a series of structured test cases covering all major functional areas. All twelve test cases passed successfully:

Test Case	Description	Result
TC-01	User Registration with Valid Details	PASS
TC-02	User Login with Valid Credentials	PASS
TC-03	Language Selection and Progress Display	PASS
TC-04	Level Listing with Lock/Unlock Logic	PASS
TC-05	Vocabulary Lesson Loading	PASS
TC-06	Quiz Question Display with Timer	PASS
TC-07	Quiz Answer Submission and Scoring	PASS
TC-08	Points Awarded After Quiz Completion	PASS
TC-09	Level Completion and Next Level Unlock	PASS
TC-10	Leaderboard Ranking and Points Display	PASS
TC-11	Session Management and Logout	PASS
TC-12	Cookie-Based Preference Storage	PASS

All test cases passed successfully, confirming that the Language Learning App with Quizzes is fully functional and ready for deployment in an educational environment.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The Language Learning App with Quizzes was successfully developed as a comprehensive web-based platform that provides an interactive, gamified, and self-paced approach to language learning. The system addresses the limitations of traditional language education by offering structured vocabulary lessons, timed quiz assessments, point-based progression, and competitive leaderboard features in a single, locally deployable application.

The application supports eight languages organized into progressive levels, each combining vocabulary instruction with quiz-based assessment. Users accumulate points through successful quiz completion, which are tracked in a centralized MySQL database and reflected on a global leaderboard. The system implements secure user authentication with session management and cookie-based preference storage, demonstrating key web technology concepts in a practical, real-world context.

Testing confirmed that all modules perform reliably and accurately, meeting the defined project objectives. The modern dark-themed interface, progressive level structure, and gamification elements create an engaging learning experience that motivates consistent practice and vocabulary retention. The system is lightweight, browser-accessible, and suitable for deployment in educational institutions, language training centres, and for individual self-study.

7.2 Future Enhancements

Several enhancements can be made in the future to increase the capabilities and reach of the system:

- Mobile Application: Developing a dedicated Android and iOS application to allow learners to practice languages on smartphones with offline access to lesson content.
- Audio Pronunciation: Adding audio playback for vocabulary words and phrases to help learners develop correct pronunciation alongside reading comprehension.
- Writing and Speaking Exercises: Introducing fill-in-the-blank, sentence construction, and speech recognition exercises to complement multiple-choice quizzes.

- Spaced Repetition System: Implementing a spaced repetition algorithm to intelligently schedule vocabulary review sessions based on individual learner performance.
- More Languages and Content: Expanding the platform with additional languages, more levels per language, and topic-specific vocabulary sets such as business, travel, and academic language.
- Instructor Dashboard: Adding an admin panel for teachers to create custom quizzes, monitor student progress, and assign specific languages or levels to student groups.
- Social Features: Introducing friend lists, challenge modes, and group leaderboards to enhance the social and competitive aspects of the platform.

These future improvements will make the Language Learning App with Quizzes a more powerful, inclusive, and feature-rich platform for learners across diverse educational and professional contexts.

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APPENDIX

Appendix 1: System Pseudocode

System Initialization Logic

```
START
  Connect to MySQL database
  IF connection successful → Start server, load login page
  ELSE → Display connection error
END
```

User Login Logic

```
FUNCTION Login(email, password):
  Query users table for matching email
  IF found AND password_verify(password, hash)
    Create session: store user_id, username, points
    Set authentication cookie
    Redirect to dashboard
  ELSE display 'Invalid credentials'
END FUNCTION
```

Quiz Submission Logic

```
FUNCTION SubmitQuiz(user_id, level_id, answers[]):
  Load correct answers for level_id from Questions table
  score = count matching answers
  points = score * 10
  UPDATE users SET total_points = total_points + points
  INSERT INTO progress (user_id, level_id, score, completed=1)
  IF score >= passing_mark → unlock next level
  Return score, points
```

END FUNCTION

Leaderboard Logic

FUNCTION GetLeaderboard():

  SELECT username, total_points FROM users ORDER BY total_points DESC

  Assign rank positions 1, 2, 3 ...

  Return ranked list

END FUNCTION

Appendix 2: System Test Results Summary

Test Case	Description	Result
TC-01	User Registration with Valid Details	PASS
TC-02	User Login with Valid Credentials	PASS
TC-03	Language Selection and Progress Display	PASS
TC-04	Level Listing with Lock/Unlock Logic	PASS
TC-05	Vocabulary Lesson Loading	PASS
TC-06	Quiz Question Display with Timer	PASS
TC-07	Quiz Answer Submission and Scoring	PASS
TC-08	Points Awarded After Quiz Completion	PASS
TC-09	Level Completion and Next Level Unlock	PASS
TC-10	Leaderboard Ranking and Points Display	PASS
TC-11	Session Management and Logout	PASS
TC-12	Cookie-Based Preference Storage	PASS